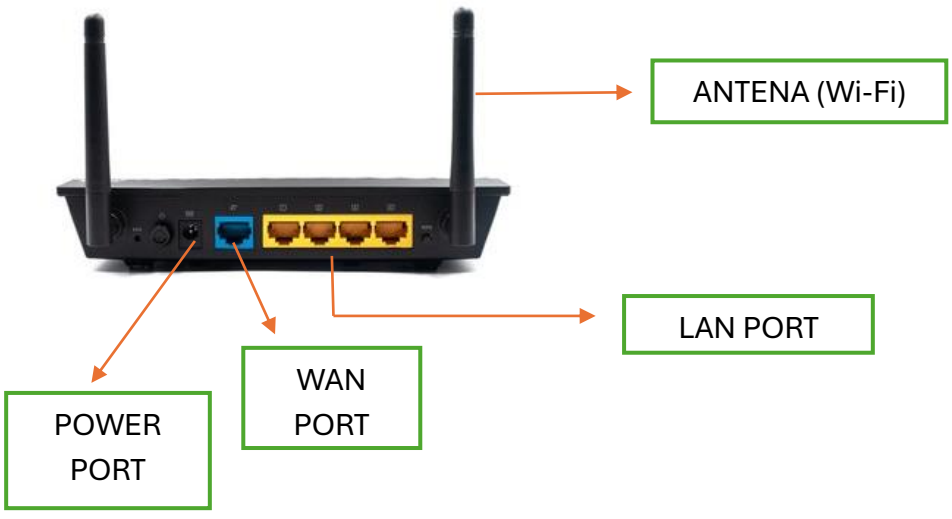
1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

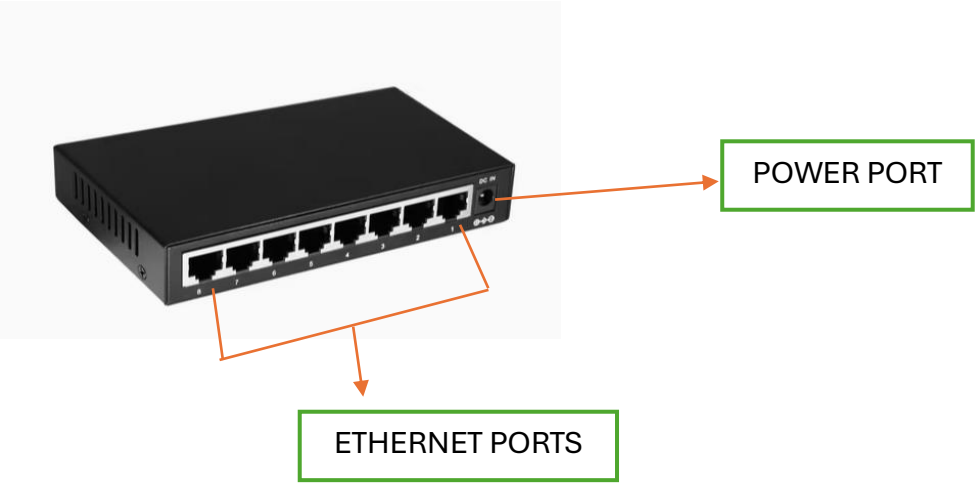
Network Device	Main Function	Typical Ports	Use in Home Wi-Fi
Switch	Connects multiple devices within a LAN and forwards data using MAC addresses.	4, 8, 16, or 24 Ethernet (RJ45) ports	Connects computers, printers, and smart TVs using Ethernet cables.
Router	Connects different networks and directs data between the home network and the internet.	1 WAN port and 2–4 LAN ports (RJ45)	Connects home devices to the internet and manages network traffic.
Hub	Connects multiple devices and broadcasts received data to all ports.	4, 8, or 16 Ethernet (RJ45) ports	Rarely used today because switches are more efficient.
Firewall	Protects the network by filtering incoming and outgoing traffic based on security rules.	WAN/LAN Ethernet ports; commonly filters TCP/UDP ports such as 80, 443, and 53	Blocks unauthorized access and protects home devices from network threats.
Access Point (AP)	Provides wireless connectivity to devices within a LAN.	Usually 1 Ethernet (RJ45) port, sometimes additional LAN ports	Allows phones, laptops, and tablets to connect through Wi-Fi.

2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

ROUTER



SWITCH



ACCESS POINT



3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

Example: Instagram App

1. **Access Point:** Connects my smartphone to the Wi-Fi network when I open Instagram.
2. **Router:** Sends my requests to Instagram's servers through the internet and receives photos, videos, and messages.
3. **Firewall:** Protects my network by filtering potentially harmful or unauthorized traffic.
4. **Conclusion:** These devices work together to provide a secure and smooth Instagram experience.

4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.

College Wi-Fi Problem and Solution

Possible Reason: The college Wi-Fi may be slow because too many students are connected to a single access point, exceeding its capacity.

Solution: Upgrade to a modern Wi-Fi 6 access point that supports more connected devices and handles network traffic efficiently.

Benefits:

1. Faster and more stable internet connections.
2. Better performance when many students are connected.
3. Fewer disconnections caused by network congestion.

Conclusion: Upgrading the access point can improve Wi-Fi speed and stability. Additional access points may be needed if poor coverage is also a problem.